

How the Proof Works

For the global three-dimensional Navier–Stokes regularity problem, the fluid obeys

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0.$$

A vortex tube inside this flow can be caught in a strain that pulls along its axis. Because $\nabla \cdot u = 0$, the volume of each fluid parcel is preserved. The tube lengthens, its cross-section contracts, and the rotating core narrows.

The tube carries its rotation in the vorticity $\omega = \nabla \times u$, which evolves by

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega.$$

The term $(\omega \cdot \nabla)u$ is the stretching of that rotation. When the strain is aligned with the vortex, $\omega \cdot ((\omega \cdot \nabla)u) > 0$, so this contribution drives $|\omega|$ upward inside the narrowing core. Because ω is already one spatial derivative of u , the velocity now changes more sharply across the core: its gradient rises.

Viscosity enters through $\nu \Delta \omega$, whose strength also depends on the width of the core. For $\lambda > 1$, shrink space and change velocity, pressure, and time with it:

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

A core of width r , velocity scale U , and duration τ becomes a core of width r/λ , velocity scale λU , and duration τ/λ^2 . Applied to the equation, this change gives

$$\begin{aligned} \partial_t u_\lambda + (u_\lambda \cdot \nabla)u_\lambda + \nabla p_\lambda - \nu \Delta u_\lambda \\ = \lambda^3 [\partial_t u + (u \cdot \nabla)u + \nabla p - \nu \Delta u](\lambda x, \lambda^2 t) = 0. \end{aligned}$$

Every term has acquired the factor λ^3 , so nonlinear sharpening and viscous diffusion have intensified in equal proportion: narrowing the core has not given viscosity an advantage.

The velocity scale grew by λ , but the occupied volume fell by λ^{-3} . Hence

$$\|u_\lambda(\cdot, t)\|_2^2 = \lambda^{-1} \|u(\cdot, \lambda^2 t)\|_2^2.$$

The gradients can grow while the kinetic energy falls.

Across the homogeneous Sobolev scale, this contraction gives

$$\widehat{u_\lambda}(\xi, t) = \lambda^{-2} \widehat{u}(\xi/\lambda, \lambda^2 t), \quad \|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-\frac{1}{2}} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}.$$

At $s = \frac{1}{2}$, the exponent vanishes, so the width of the core changes without changing its $\dot{H}^{1/2}$ size. Below that height the norm decreases; above it the norm increases. This makes $s = \frac{1}{2}$ the critical height.

Now assume one evolving vortex repeats the full scale change, reducing its width by a fixed factor $0 < \rho < 1$ at each turn. Then

$$\begin{aligned} r_j &= \rho^j r_0, & U_j &= \rho^{-j} U_0, & \tau_j &= \rho^{2j} \tau_0, \\ \sum_{j=0}^{\infty} \tau_j &= \tau_0 \sum_{j=0}^{\infty} \rho^{2j} = \frac{\tau_0}{1 - \rho^2} < \infty. \end{aligned}$$

Under this hypothesis, infinitely many turns fit before finite time while $U_j \rightarrow \infty$: a Zeno cascade compatible with finite duration and falling kinetic energy.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

What you'd expect to see in this pattern, if a singularity were to form, as you go up the time derivatives, you'd expect to see less spatial regularity, not more.

The equation moves in the other direction. Its first three time derivatives are

$$\begin{aligned} \partial_t u &= \nu \Delta u - (u \cdot \nabla) u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla) u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla) u - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

The pattern is

$$\partial_t^{n+1} u = \nu \Delta \partial_t^n u - \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u - \nabla \partial_t^n p.$$

Set

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p,$$

Taking the divergence and using incompressibility gives the pressure at that order:

$$-\Delta Q_n = \sum_{a=0}^n \binom{n}{a} \partial_i \partial_j ((V_a)_i (V_{n-a})_j).$$

The line to keep in view is $V_{n+1} = \nu \Delta V_n - \dots$. Moving one step up in time brings in two spatial derivatives of the response below it. Transport and pressure stay attached, but viscosity supplies the link from temporal order to spatial Sobolev order.

At the critical height, set

$$\Lambda = (-\Delta)^{1/2}, \quad E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t).$$

If $\mathcal{P}_s$ denotes the complete transport–pressure contribution at height s , the energy identity is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

Starting from $H = E_{1/2}$ and repeating that identity gives

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

The first term has reached spatial height $n + \frac{1}{2}$; the sum carries every transport–pressure term generated on the way. This is the opposite of the blow-up picture. Rising through that temporal sequence was supposed to reveal less spatial regularity. The equation exposes higher Sobolev height instead. As you do that, you get higher order Sobolev embeddings:

$$s > k + \frac{3}{2} \implies H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3), \quad \bigcap_{s < \infty} H^s(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Seeing a higher energy in the identity is not enough. It has to remain finite all the way to T_* . Fix smooth divergence-free finite-energy data u_0 , viscosity $\nu > 0$, and its maximal pressure-complete solution (u, p) on $[0, T_*)$. Assume $T_* < \infty$. For finite N and M , collect the adjacent rows in

$$\mathfrak{R}_{N,M}(u; T) := \sum_{n=0}^N \left(\|V_n\|_{L^2(0,T;H^{M+1})}^2 + \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2 \right).$$

Adding the corresponding Q_n and ∇Q_n coordinates gives the finite pressure-complete rectangle $\mathcal{R}_{N,M}[u_0; T]$. The manuscript's whole-terminal theorem bounds each adjacent pair on the full interval:

$$V_n \in L^2(0, T_*; H^{M+1}), \quad V_{n+1} \in L^2(0, T_*; H^{M-1}), \quad 0 \leq n \leq N.$$

Restriction to $(0, T)$ can only decrease those norms, and there are only finitely many rows, so

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty.$$

The bound may depend on N and M . No uniform constant over all orders is needed. At $n = 0$, for every finite m , the adjacent pair gives

$$u \in L^2(0, T_*; H^{m+1}), \quad \partial_t u \in L^2(0, T_*; H^{m-1}),$$

The Lions–Magenes trace theorem applied to $H^{m+1} \hookrightarrow H^m \hookrightarrow H^{m-1}$, gives a unique trace at the endpoint:

$$u \in C([0, T_*]; H^m) \quad \text{for every finite } m.$$

Increasing N or M does not create a new solution. Every rectangle is a finite part of the maximal history from u_0 , so overlapping rectangles agree on every shared coordinate. Their projective limit is

$$\mathcal{I}[u_0] := \varprojlim_{N, M < \infty} \mathcal{R}_{N,M}[u_0; T_*], \quad u \in \bigcap_{r, m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)).$$

The intersection is not one bound over infinitely many orders. Each coordinate was bounded at a finite (N, M) ; compatibility says that all of those finite bounds belong to one velocity–pressure history.

The zeroth temporal row, $\mathcal{S}_0 := \pi_0 \mathcal{I}[u_0]$, now has one terminal value at every finite spatial height:

$$u_* := u(T_*) \in \bigcap_{m < \infty} H_\sigma^m(\mathbb{R}^3) = H_\sigma^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The recurrence and pressure equation recover every finite coordinate of the terminal jet from u_* . The uniform H^3 row gives a local existence time $\delta_0 > 0$ that works for u_* and for every sufficiently late slice $u(t_0)$. Choose $t_0 < T_*$ with $T_* - t_0 < \delta_0/2$, and restart from $u(t_0)$. Uniqueness makes this restart identical to the original solution on $[t_0, T_*)$, so it reaches T_* with value u_* . To the right of T_* , uniqueness identifies it with the solution restarted from u_* . The two restarts join into one smooth pressure-complete solution beyond T_* , contradicting maximality. Thus $T_* = \infty$.

Once $V_r \in C_t H_x^m$ is available for every finite r and m , any finite downstream estimate comes from choosing

$$m > k + \frac{3}{2} - \frac{3}{p}, \quad 2 \leq p \leq \infty,$$

which gives, on every finite time interval,

$$V_r \in C_t H_x^m \hookrightarrow L_t^q W_x^{k,p}, \quad 1 \leq q \leq \infty.$$

$$u_0 \longrightarrow \{(V_n, Q_n)\}_{n \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N, M < \infty} \longrightarrow \mathcal{I}[u_0] \longrightarrow \mathcal{S}_0 \longrightarrow u_* \in H^\infty \longrightarrow T_* = \infty.$$